

Synchrony-based Depression Score Aggregation from Single-Modality Models

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ABSTRACT

Depression is a mental health problem that affects human mood and the ability to function properly. Currently, the assessment is mainly based on subjective questionnaires and clinical opinions. This work aims to infer the depression level through analysing multimodal data, as a tool to support medical experts and patients in depression screening and diagnosis. We introduce a fusion method to estimate the Beck Depression Inventory II scores using multiple inputs: facial images, video-based blood volume pulse signals, and speech data. Each modality has its own regression model, based on the ResNet-50 architecture. Our approach leverages the synchrony between regression scores of these models to produce the fusion values. Specifically, we calculate the Pearson correlation coefficient and the dynamic time warping distance between sliding windows of the score sequences to find the optimal segments for fusion. We evaluate our method on the dataset of the fourth Audio-Visual Emotion Recognition Challenge (AVEC 2014). We achieve a Mean Absolute Error of 6.08 and a Root Mean Squared Error of 8.60, which are lower than those of each single-modality model.

CCS CONCEPTS

• Computing methodologies → Ensemble methods.

KEYWORDS

multimodal data; ensemble methods; wellbeing

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1 INTRODUCTION

It is estimated that major depressive disorder (commonly known as depression or MDD) affects 3.8% of the worldwide population¹. Depression screening, diagnosis, and treatment outcome monitoring are based primarily on questionnaires and interviews [2] [14]. These protocols require dedicated training programs for medical personnel and routine evaluation of patient well-being [18]. Recent developments in human behaviour recognition may become a useful tool for automatic depression monitoring, and potentially facilitate an objective evaluation. Automatic depression recognition systems have been developed, using speech signals [21], videos [17], wearable sensors [12], and natural language processing [23]. Multimodal approaches that combined speaking behaviour, eye activity, and head pose features showed their effectiveness to distinguish depression and non-depression [1]. To compare the relative merits of audio and video data in emotion recognition and depression level estimation, Valstar *et al.* [19] introduced the fourth Audio-Visual Emotion Recognition Challenge (AVEC 2014). This challenge provided visual and audio data from participants, as well as the Beck Depression Inventory II (BDI-II) scores [3] to indicate the severity of depression.

In this work, we study score aggregation methods to estimate the depression level. These methods summarize the output scores of multiple regression models. This approach provides a flexible implementation: the end-user can decide to use any sub-set of the models. As a motivation example, Figure 1 shows the scores of the models trained on three modalities: videos, remote photoplethysmography (rPPG) signals, and voice. We observed that a modality may have a better result than others in some videos. Hence, we expect the combination of these scores might yield an improved estimation of the depression level (i.e., the BDI-II score [3]).

To calculate the fusion scores, we leverage the synchrony between the score sequences produced by the single-modality models. Synchrony measures have been used in multimedia analysis, such as monologue detection based on the synchrony between speech and facial movements [10] and biometric identity verification based on the correspondence between acoustic and visual speech [4]. In our work, for each pair of score sequences, we segment them

¹Depression: <https://www.who.int/news-room/fact-sheets/detail/depression>

with a fixed-length sliding window. We then implement two variants of our fusion technique: one using the Pearson correlation coefficient [8] to locate the segments with the highest synchrony and another using the dynamic time warping (DTW) distance [16]. Lastly, we average the values of the most synchronous segments to form the final fusion score. We perform experiments on the AVEC 2014 dataset [19] in order to show that the proposed fusion method produces better results than the outputs of each single-modality model, in terms of the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics.

2 RELATED WORK

Human emotions can be perceived via multiple modalities, such as speech, facial expressions, and physiological signals. There are three generic fusion techniques that integrate features, scores, or decisions [11]. Lingenfeller *et al.* [13] investigated 16 fusion methods for recognizing emotions: several basic emotions and neutral. They showed that multimodal data provided more discriminative information for various emotion recognition tasks. The decision fusion methods included majority voting, algebraic combination rules, and sequential combination of individual decisions. On detecting depression, Joshi *et al.* [11] studied fusion methods on audio-video features at different levels: feature concatenation; combination of different scores such as probability estimates and likelihoods; and decision fusion based on boolean operators. Alghowinem *et al.* [1] explored feature and decision fusion methods for head poses, eye movements, and speech signals to detect depression. Their results showed that fusing all modalities offered a remarkable improvement in comparison to uni-modal systems. The aforementioned articles focused on detecting depression as a classification problem while our work utilize multiple regression models to estimate the depression level.

In order to assess the level of depression, Meng *et al.* [15] applied a weighted sum rule to combine the outputs of regression models trained on visual and vocal features. The weights were derived using the minimum classification error criterion. Similarly, Williamson *et al.* [20] also utilized a weighted score fusion scheme. They used the cross-correlation between the features (voice and facial action units) and the BDI-II scores to compute the optimal weights. Instead of analyzing the training data to obtain the optimal combination rules, we leverage the synchrony of decisions to combine them. This approach makes the decision fusion step independent from the training process of each single-modality model. Furthermore, it can be deployed with any subset of the modalities.

3 SYNCHRONY-BASED SCORE AGGREGATION

To assess the depression level, we analyse data from three modalities: facial images, audio, and rPPG signals. Each modality has its own regression model based on the ResNet-50 architecture [9] to estimate the BDI-II scores [3]. Figure 2 displays several modalities used to estimate the depression level: video frames (facial images), rPPG signals, and speech data. We hypothesize that the best fusion score can be obtained at the segments with the highest synchrony metrics.

3.1 Single-Modality Models

We utilize a ResNet-50 architecture [9] with two fully connected (FC) layers of 2048 neurons and a regression layer of 128 to estimate the BDI-II score [3] as the depression level. We initialize the network with pre-trained weights on the ImageNet dataset [7]. Each modality has its own ResNet-50 as the single-modality model.

For video data, we perform the pre-processing step: aligning and cropping the facial region in every video frame. We apply a Multi-task Cascade Convolutional Network [22] to detect and align the faces, using five fiducial landmarks. These aligned-and-cropped video frames are the inputs of the ResNet-50. The output of the video model was a depression score for each video frame.

To acquire the rPPG signals, we measure the pulsatile signals of the blood volume from the faces unsupervised [5]. We then segment the rPPG signals into 6-second windows with a sliding step of 0.33 second. The duration is equivalent to 10 video frames with a frame rate of 30 frames per second. We transform each window into a spectrogram, filtering out those frequencies not present in the pulse (i.e., 0.5-3.5Hz) and use it as the input to the ResNet-50.

For voice data, the audio samples are converted into single-channel signals and down-sampled to the sampling rate of 16 kHz. The audio data is split into multiple 3-second segments with a step of 0.25 second. A Hamming window with 25ms width and 10ms step is applied to the audio segments [6]. Then, for each segment, we compute a spectrogram of size 512×300 , where 512 is the size of the spectrum and 300 is the number of frames. Then the spectrogram is normalized by performing mean and variance normalization.

The score sequences produced by different single-modality models may have different sizes. Hence, we applied interpolation and resampling to make all of them have the same number of samples.

3.2 Synchrony-based Aggregation

For every pair of single-modality models, we quantify their synchrony by sliding a fixed-length window over two score sequences. For each window, we calculate the Pearson correlation coefficient [8] and the DTW distance [16] between two segments. The Pearson correlation coefficient indicates the linear relationship of two time series sequences as a number from -1 (negatively-correlated) to 1 (perfectly-correlated). The DTW technique is equivalent to minimizing the Euclidean distance between two time series sequences over all possible temporal alignments. The Pearson correlation coefficient that is close to 1 is a sign of high synchrony. On the other hand, a small DTW distance shows a possibility that the two segments have a high synchrony.

After calculating the pairwise synchrony metrics for all windowed segments, we find the pair that has the highest synchrony. Then, we compute the mean of the scores in these two segments to obtain the final result. For comparison, we calculate two error metrics between the fusion results and the ground-truth BDI-II scores. Since the window length may affect the fusion scores, we vary this parameter to find the optimal value for both synchrony metrics.

4 EXPERIMENTS AND ANALYSIS

Our experiments were performed on a desktop with an AMD Ryzen 7 5800 8-core processor and an NVIDIA GeForce RTX 3060. We

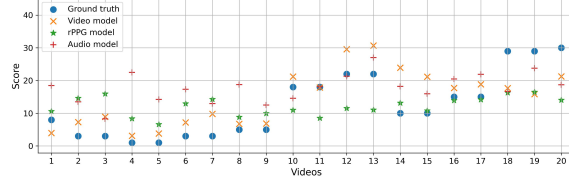


Figure 1: Depression scores in 20 videos [19], estimated by three single-modality models: videos, remote Photoplethysmography (rPPG) signals, and audio data

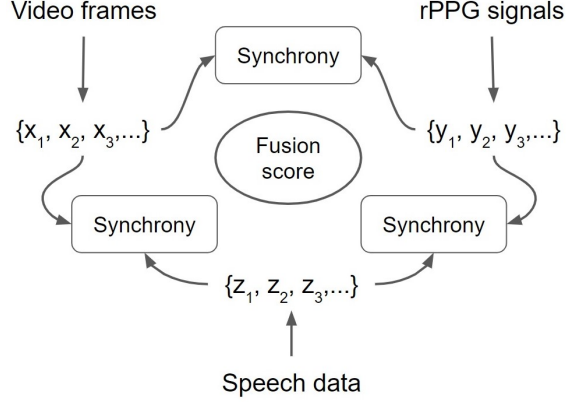


Figure 2: Synchrony-based score aggregation from single-modality models

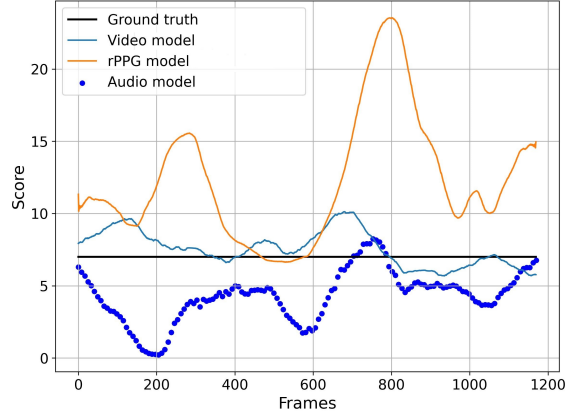


Figure 3: Score predictions of models trained on different modalities from one subject

used Python 3.8 as the programming language and PyTorch 1.11.0 as the machine learning framework. We evaluated our approach in the audiovisual dataset in the AVEC2014 challenge [19]. The cost of each regression model is calculated with a mean average error function (MAE). The dataset contained 300 videos of subjects speaking in front of a camera. The participants answered 21 questions about how they had been feeling in the previous week. Each question had been scored on a scale value from 0 to 3 (higher scores indicated

Table 1: Mean Average Error (MAE) and Root Mean Squared Error (RMSE) of single-modality models and our fusion schemes

	MAE	RMSE
Video (facial images, aligned and cropped)	6.22	9.22
rPPG (spectrogram)	9.00	11.00
Audio (spectrogram)	7.96	10.16
Fusion (Pearson correlation coefficient)	7.02	8.82
Fusion (Dynamic time warping distance)	6.08	8.60

more severe depressive symptoms). The responses were used to compute the BDI-II scores [3] indicating the level of depression. The severity of depression could be categorized as: Normal mood ups and downs (0-10), Mild mood disturbance (11-16), Borderline clinical depression (17-20), Moderate depression (21-30), Severe depression (31-40), and Extreme depression (40-63). In this work, we considered Northwind and Freeform videos as a joined task. For comparison, we provided two performance metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), calculated between our estimated values and the ground-truth BDI-II scores.

Table 1 summarizes our results with the best error metrics using the Pearson’s correlation coefficient (window length of 128 samples) and the DTW distance (window length of 32 samples). When we utilized the Pearson’s correlation coefficient as an indicator to find the segments with the highest synchrony, we were able to estimate the fusion depression scores at a lower RMSE metric than that of the single-modality models. However, the MAE metric of this fusion scheme was higher than that of the video model. If the DTW distance was used to measure the synchrony, the fusion scores were better in both metrics.

Figure 4 visualizes the fusion results when the window length is varied: 16, 32, 64, and 128 samples. The step length is one sample. With the variant based on the Pearson’s correlation coefficient, the optimal window length is 128, where the MAE and RMSE metrics are lowest: 7.02 and 8.82, respectively. With the other variant that utilizes the DTW distance, the best window size is 32 if we prefer the lower MAE (6.08). Otherwise, a window size of 64 gives the best RMSE metric: 8.51.

The results show that our fusion method can enhance the estimated depression level. The score segments in synchrony with each other represent the agreement of an expert committee. Hence, if the outputs of all individual models become better, our approach may improve the fusion results.

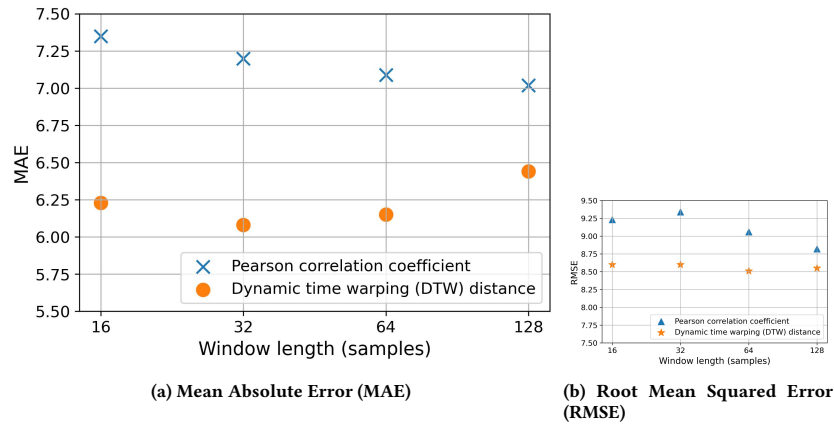


Figure 4: Error metrics of the fusion method with varying window lengths

5 CONCLUSION

We present a fusion method to aggregate depression scores produced by single-modality models. For each modality, we independently train a regression model based on the ResNet-50 [9] architecture. Our approach leverages the synchrony of the score sequences produced by video-, rPPG-, and audio-based models. The evaluation in the AVEC2014 dataset shows that the fusion scores are better than those of each single-modality model, in terms of MAE and RMSE metrics. Since our method relies only on the outputs of single-modality models, it can integrate depression scores from an arbitrary set of single-modality models. In addition, visualizing the score sequences with their synchrony may assist the medical experts in depression screening and diagnosis.

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